

Session 18: JavaScript Control Flow & Operators

Session Objective

Help learners control program flow using **conditions and logical thinking**, while understanding different types of operators and how JavaScript handles **type conversion**.

Duration

3 Hours

Instructor Guidelines

- Follow Agenda: Please adhere to the provided session agenda to ensure effective and consistent training delivery. Use the provided training resources such as slide decks and code samples to teach in the class. Use references, wherever provided to learn and prepare with factual content before conducting the session.
- Check Audio Equipment: Verify that your microphone and speakers/headset are functioning properly before starting the session. Please use a good quality microphone and completely avoid using bluetooth earbuds as they employ noise-cancellation which reduces audio quality.
- Camera Usage: Please ensure your camera is on for at least the first 15 minutes of the session.
- Power Backup: Please ensure you have a power backup solution ready to prevent interruptions due to power outages.
- Internet Bandwidth: Please confirm that your internet connection is fast enough to handle live-streaming audio, video, and screen sharing.
- Code/Query Testing: Run through your code/queries before the session to prevent any disruptions.
- Hands-On Coding: During interactive coding segments, explain steps clearly and frequently check in with students to confirm their understanding.
- Technical Support: If you encounter issues with Zoom, contact the session moderator or the learning advisor for help.

Agenda

1. Session Opening

- Greet learners and take attendance
- Recap previous session (variables, data types, operators)
- Set expectation: today = **decision making in code**

- Ask learners to open VS Code

2. Assignment Operators

- Explain assignment concept (=)
- Introduce compound operators:
 - +=, -=, *=, /=
- Demonstrate simple examples
- Show real usage (updating values)

3. Logical Operators

- Explain need for combining conditions
- Introduce:
 - && (AND)
 - || (OR)
 - ! (NOT)
- Demonstrate using real-life conditions:
 - login validation
 - age + eligibility
- Keep examples relatable

4. Unary Operators

- Explain operators that work on single value
- Introduce:
 - ++ (increment)
 - -- (decrement)
- Show prefix vs postfix behavior (basic only)
- Keep examples simple

5. Type Conversion (Important Concept)

- Explain why type conversion happens
- Introduce:
 - implicit conversion (automatic)
 - explicit conversion (manual)
- Demonstrate:
 - Number(), String()
- Show common confusion cases:
 - "5" + 2
 - "5" - 2
- Emphasize awareness over memorization

6. Break

- Ask learners to step away, hydrate, stretch
- Resume on time

7. If/Else Statements (Core)

- Explain conditional execution
- Teach structure:

```
if (condition) {  
    // code  
} else {  
    // code  
}
```

- Demonstrate:
 - even/odd check
 - pass/fail logic

8. Nested Conditionals

- Explain conditions inside conditions
- Demonstrate simple example:
 - login → then role check
- Warn against deep nesting (basic mention)

9. If-Else-If Ladder

- Explain multiple conditions scenario
- Teach structure:

```
if (condition1) {  
} else if (condition2) {  
} else {  
}
```

- Demonstrate:
 - grading system
 - pricing logic

10. Switch Statement

- Explain alternative to multiple if-else
- Teach structure:

```
switch(value) {  
    case x:  
        break;  
    default:  
}
```

- Demonstrate simple example:

- day of week
 - menu selection
- Keep use-case limited

11. Practice Exercises (Hands-On)

- Ask learners to:
 - build a grading system
 - check login eligibility
 - use switch for menu selection
- Ensure they type and run code

12. Doubt Resolution

- Address:
 - if vs switch confusion
 - logical operators misuse
 - type conversion confusion